

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. Computer Engineering Sem - II

University Examination, 2010-11

CE 712 Software Project Management & Quality Assurance (SPM & QA)

Date: 18.05.11, Wednesday Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

- Instructions:** (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work to be done in the last page of main supplementary, don't write anything on the question paper.

SECTION-I

Q-1 Answer the following questions.

1. Many software projects are so large that a no. of software development teams must be formed. Recommend a management structure for coordinating multiple teams. What are the potential problems that can arise? 3
2. Explain how project manager would carry out the risk analysis? What would be the outcome of the risk analysis? How would the outcome of your analysis be used to manage the risk? 3
3. Why is Project Management important to the software development process? What is the critical distinction between a milestones and deliverables? 3
4. "Software quality managers are responsible for three kinds of activities". Name and explain them in brief. 2

Q-2

- [A] Compare the relative advantage of using the iterative waterfall model and the spiral model of software development. Explain with the help of examples, the type of problems for which you would adopt the waterfall model of software development, and the type of problems for which you would adopt the spiral model. 4
- [B] "Reliability normally should take precedence over efficiency". Is this statement true? Justify your answer. 4

OR

- [B] Discuss merits and demerits of ISO 9001 and SEI CMM certification. 4
- [C] The Project which is handled by you is in *RED*, customer has given you 4 weeks to show improvement, and still nothing has worked out. The Customer again gives 3 weeks to improve and you do your best to improve, but nothing works out. Now the customer is not at all happy, How to convince this customer? 4

Q-3

- [A] What are the four attributes which all software products should have? Explain each of them in brief. Suggest four other attributes which may be significant. 4
- [B] Suppose the project has a very tight schedule. Suggest two ways in which productivity could be improved to help bring this project in on schedule. Discuss how each of the methods you describe actually improves productivity. 4

OR

- [B] List the major activities of the software project manager. List soft skills required by the project manager. 4
- [C] What is Function Point Analysis? Compute the FP value for a project with the mentioned information domain. 4

No. of user I/P	:32
No. of user O/P	:60
No. of user enquiries	:24
No. of files	:08
No. of External I/F	:02

SECTION-II

Q-4

1. What is the difference between Audit and Formal Technical Review (FTR)? Can their function be folded in to the review? What are the pros and cons? 3
2. Is it true that "High quality software process should lead to high quality software products? Justify. 4
3. Is it true that a software product can always be developed faster by having a larger development team of competent software engineers? Justify your answer. 4

Q-5

- [A] What is the difference between a revision and a version? What do you mean by version control? Why is version control required? How can version control be achieved? 4
- [B] How is McCabe's Cyclomatic complexity metric useful? Analyze the usefulness of McCabe's Cyclomatic complexity. 4
- [C] If an organization does not document its quality system, what problems would it face? What do you understand by Total Quality management (TQM)? What are the advantages of TQM? 4

OR

- [A] What is the difference between code inspection and code walkthrough? Briefly explain the types of errors that are expected to be detected in each. Why is it advantageous to detect defects during code inspection than to detect these during testing? Briefly explain how code inspection can be carried out. 4
- [B] What do you mean by measure, measurement and metric? How they are related to each other? What is difference between Product and Process Quality metrics? 4
- [C] You are asked by your project manager to deliver software to a schedule, which you know can only be met by working overtime (unpaid). All team members are newly married & have young children. Discuss whether you should accept this demand from your team to give their time to the organization rather than to their families. What factor might be significant in your decision? 4

Q-6

- [A] You are implementing a SAP project. Design is completed and development started and at that time you found that there is major problem with the system. It would take again time and customer is not happy with that, in that case how do you convince the customer? 4

OR

- [A] What are the fundamental tasks that must be included in SQA planning? 4
- [B] What are the advantages and disadvantages of LOC and COCOMO metrics? 4
- [C] What is the level of criticality in Project management for a Project which is started in maintenance Phase? Which phase of Project management theory must be more focused for such kind of projects? 4

OR

- [C] What can you make out as a newly approached client, if the approaching software company is currently at CMM Level 4 and has an ISO 9001 certification. 4